

# Shuhei Watanabe

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## Education

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- Oct 2020 – Oct 2023    **University of Freiburg**, Freiburg, Germany.  
MSc in Computer Science. Supervisor: Prof. Frank Hutter.  
Overall GPA: 1.1/5.0 (1.0 is the best grade).
- Apr 2014 – Mar 2020    **The University of Tokyo**, Tokyo, Japan.  
BSc in Systems Innovation (Apr 2016 ~), Liberal arts (~ Mar 2016).  
**Ranked 1st** out of 37 students by GPA.

## Employment

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- Jan 2026 – Present    **SB Intuitions Corp.**, Tokyo, Japan. (Robotics Senior Research Scientist)
- Jun 2024 – Dec 2025    **National Institute of Advanced Industrial Science and Technology (AIST)**, Tokyo, Japan. (Part Time Visiting Researcher at Social Intelligence Research (SIR) Team)
- Oct 2023 – Dec 2025    **Preferred Networks Inc.**, Tokyo, Japan. (Research Engineer)  
Core Optuna developer. Delivered significant speedup of TPESampler (300x), the default sampler in Optuna, and GPSampler (10x faster than BoTorch). Led the development of GPSampler and its extensions. Deployed Optuna in materials & physics simulation applications.
- Dec 2020 – Sep 2023    **Machine Learning Lab**, Freiburg, Germany.  
Development of Auto-PyTorch, an AutoML tool.
- Sep 2018 – Sep 2020    **AIST**, Tokyo, Japan. (Full Time Technical Staff at SIR Team)  
Co-first authored “Warm Starting CMA-ES for Hyperparameter Optimization” (AAAI’21). Conducted a large-scale experiment on a cluster.

## Selected Awards / Honors

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- **KDD Best Reviewer** (Research Track), ACM SIGKDD 2026.
- **IJCAI-AIJ 2023 Travel and Accessibility Grant Program** (1,000 USD).
- **ELIZA MSc Scholarship** (12,000 EUR, 4 students in the University of Freiburg).
- **ITO Foundation for International Education Exchange** (48,000 USD, AR: 6.7%).
- **Hatakeyama Award from the Japan Society of Mechanical Engineers** (top 1.5% ( $\approx 5/340$ ) grades in the mechanical-related faculties at the University of Tokyo).
- **PRMU 2018 Yearly Research Encouragement Award** for “*Speed Up of Hyper-Parameter Tuning with Nelder-Mead Method by Parallel Computing*”. (AR:  $3/170=1.8\%$ )

## Selected Publications

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See [my website](#) for the full publication list. The total citation count is **1800+** as of October 2026 on Google Scholar. The acceptance rate of IJCAI’23 was about 14%.

1. ♣ Y. Ozaki, ♣ **S. Watanabe**, T. Yanase (2026). OptunaHub: A Platform for Black-Box Optimization. Journal of Machine Learning Research (JMLR).

2. **S. Watanabe**, A. Bansal, F. Hutter (2023). PED-ANOVA: Efficiently Quantifying Hyperparameter Importance in Arbitrary Subspaces. International Joint Conference on Artificial Intelligence (IJCAI). (**The default algorithm in Optuna**)
3. **S. Watanabe** (2023). Tree-Structured Parzen Estimator: Understanding Its Algorithm Components and Their Roles for Better Empirical Performance. arXiv:2304.11127. (**900+ citations on Google Scholar**, the first detailed paper about the Optuna default sampler)

## Mentoring & Supervision

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| Jun 2024 – Mar 2026 | <b>Chisa Mori</b> , MSc Student, AIST.<br>Parallel coordinate plots for multi-objective problems (Accepted to iV).                                                                                                                                  |
| Jul 2024 – Present  | <b>Kaito Baba</b> , MSc Student, Preferred Networks Inc.<br>1. Development of constrained optimization for the Gaussian process-based sampler (single-objective, multi-objective).<br>2. PED-ANOVA for conditional search spaces (Accepted to KDD). |
| Aug 2025 – Present  | <b>Kaichi Irie</b> , MSc Student, Preferred Networks Inc. & AIST.<br>Development of parallel processing in the Gaussian process-based sampler (article, workshop paper).                                                                            |

## Technical Highlights

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- Cluster Experience (MPI, parallel programming, large-scale experiments, MOAB, Slurm)
- Physics-Based Simulation (lattice Boltzmann, numerical integration, constraint satisfaction)
- Machine Learning Understanding (deep learning, reinforcement learning)
- Software Engineering (Python, C++, PyTorch, team development)
- Strong Mathematical Background (statistics incl. measure theory, optimization)
- Applications of Bayesian Optimization (materials science & Sim2Real gap)

## Miscellaneous

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- **Japanese** (Native Language), **English** (C1, TOEFL iBT: 100), **German** (B2)
- Approx. Top 3.5% (highest) algorithm programmer in **AtCoder** mostly using C++.